

1. System Architecture & Core Capabilities

The **Emerging Defense Solutions (EDS) SOC Command Center** provides deterministic verification for LLM token decoding alongside real-time hardware digital twin telemetry. Operating under AMD SEV-SNP guest enclaves (VMPL 0), the system enforces policy constraints using the Z3 SMT logic solver while delivering hands-free operational control via a speech-to-speech assistant modeled after JARVIS with a Jamaican accent.

2. Mathematical Framework & Formal Proofs

Monad Logit Transformation: Candidate token output logits L_i are modified deterministically prior to decoding:

$$L_{\hat{i}} = L_i + \log(\Phi(v_i))$$

Where $\Phi(v_i) = 1$ if Z3 SMT evaluation returns SAT, and 0 if UNSAT (reducing candidate logit to $-\infty$ and guaranteeing $P_{\text{violation}} = 0$).

Formal Access Control Property:

$$\text{HasCUI}(v_i) \Rightarrow (\text{IsEncryptedEnclave} \text{ AND } \text{IsVerifiedSession})$$

3. Hardware & Power Requirements

Hardware Accelerator	Power Draw (kW)	Thermal Recovery Efficiency
Cerebras CS-4 Engine	28.00 kW	91% Heat Exhaust Captured
Cerebras CS-3 Engine	23.00 kW	91% Heat Exhaust Captured
NVIDIA B200 HGX Cluster	14.30 kW	91% Heat Exhaust Captured
NVIDIA H200 HGX Cluster	7.00 kW	91% Heat Exhaust Captured
AMD AI Halo Unit	0.75 kW	Direct Liquid Immersion

4. Troubleshooting & Operational Fixes

Symptom / Error	Root Cause	Resolution Strategy
Port 7870 in use	Orphaned Python server process	Execute: Stop-Process -Name 'python' -Force
ElevenLabs 401 Error	Key missing tts_write access	Generate Full Access key at elevenlabs.io
Silent Audio Playback	Browser autoplay policy block	Base64 HTML5 stream auto-plays automatically